Can RL Improve Generalization of LLM Agents? An Empirical Study

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Reinforcement fine-tuning (RFT) has shown promise for training LLM agents to perform multi-turn decision-making based on environment feedback. However, most existing evaluations remain largely in-domain—training and testing are conducted in the same environment or even on the same tasks. In real-worlddeployment, agents may operate in unseen environments with different background knowledge, observation spaces, and action interfaces. To characterize the generalization profile of RFT under such shifts, we conduct a systematic study along three axes: (1) within-environment generalization across task difficulty, (2) cross-environment transfer to unseen environments, and (3) sequential multi-environment training to quantify transfer and forgetting. Our results show that RFT generalizes well across task difficulty within an environment, but exhibits weaker transfer to unseen environments, which correlates with shifts in both semantic priors and observation/action interfaces. In contrast, sequential training yields promising downstream gains with minimal upstream forgetting, and mixture training across environments improves the overall balance. We further provide detailed analyses and deeper insights, and hope our work helps the community develop and deploy generalizable LLM agents.

1. Introduction

Reinforcement fine-tuning (RFT) has emerged as a promising post-training paradigm for improving large language model (LLM) agents on complex interactive tasks such as web navigation and software engineering (Deng et al., 2023; He et al., 2024; Jimenez et al., 2023; Merrill et al., 2026; Zan et al., 2025; Zhou et al., 2023). In RFT, an agent is trained to make a sequence of intelligent decisions that maximize task-specific objectives based on environment feedback (Mai et al., 2025; Xi et al., 2025c). By optimizing for long-horizon outcomes, RFT is often associated with stronger agentic behaviors such as instruction following (Bai et al., 2022b; Ouyang et al., 2022a), planning (Shinn et al., 2023; Yao et al., 2022b), reasoning (Lightman et al., 2023; Uesato et al., 2022), and tool use (Li et al., 2023, 2025b).

Despite rapid progress, most empirical evidence for RFT focuses on in-domain evaluation, where training and testing are conducted within the same environment—and often even on closely overlapping tasks (Liu et al., 2023; Zhou et al., 2023). In real-world deployment, however, agents frequently encounter unseen environments that differ not only in task instances, but also in background knowledge, observation spaces, and action interfaces (Ruan et al., 2023; Xi et al., 2025a). Such shifts can fundamentally reshape the interaction dynamics faced by agents, including which observations are informative, what actions are feasible, and how failures can be corrected. This raises a practical and

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important research question: do the improvements brought by RFT generalize beyond the training distribution?

To bridge this gap, we conduct a systematic study to explore how RFT impacts the generalization and transferability of LLM agents across three axes (Figure 1). First, we investigate task-level generalization under the same environment by reviewing varying difficulty performance when RFT on a specific subset of tasks (Section 4); Second, we examine environment-level generalization to assess whether RFT maintains efficacy in unseen environments, where agents must navigate shifts in background knowledge alongside changes in observation and action spaces (Section 5). Third, we analyze sequential training across environments to characterize the dynamics of transfer and forgetting, comparing this approach against joint training on environmental mixtures (Section 6). Together, these three axes provide a comprehensive, systematic framework for understanding RFT-driven generalization and transfer in LLM agents.

Figure 1 | An overview of three axes we study.

Table 1 | Characteristics of each environment we study. The columns "DEN.", "VAL.", "ACT.", "KNWL." and "STR." indicate whether the rewards are *dense*, action validation is *strict*, *valid action lists* are provided per step, *world knowledge* is required, and observations are *structured*, respectively. The environments vary along these dimensions.

Environment	Types	DEN.	VAL.	ACT.	KNWL.	STR.
WebShop	*Web*	✔	✗	✗	✔	✔
SearchQA	*Search*	✗	✔	✗	✔	✔
TextCraft	*Game*	✗	✔	✗	✗	✗
AlfWorld	*Household*	✗	✔	✗	✗	✗
BabyAI	*Embodied*	✔	✗	✔	✗	✗

Our analysis covers five agentic environments that vary in background and core properties, as shown in Table 1 and Table 4. We highlight several key findings: (1) Intra-environment generalization, within a consistent environment, RFT-trained agents exhibit significant generalization, and easy-to-hard curriculum learning further boosts performance gains. (2) Inter-environment sensitivity, while RFT enhances agentic capabilities, its generality to unseen environments exhibits fluctuations. Our analysis reveals that this across-environment generalization is sensitive to the required prior knowledge, observation spaces, and action spaces. (3) Multi-environment training. Employing multi-environment training strategies substantially enhances generalization: sequential RFT enables effective transfer to new environments while maintaining performance on upstream ones, and mix RFT across multiple environments achieves consistently strong results overall. We further provide insights that shed light on the mechanisms underlying these behaviors. Overall, we hope these findings inform the development of more generalizable LLM agents and their practical deployment in real-world settings.

2. Related Work

2.1. RL for LLM Agents

Reinforcement learning (RL) has been widely adopted in LLM training, with paradigms such as reinforcement learning from human feedback (RLHF) (Bai et al., 2022a; Christiano et al., 2017; Ouyang et al., 2022b) and reinforcement learning from verifiable rewards (RLVR) (DeepSeek-AI, 2025; Rafailov et al., 2023; Schulman et al., 2017) demonstrating strong effectiveness in improving instruction following, reasoning accuracy, and behavioral alignment (Guo et al., 2025; Mu et al., 2024; Ren et al., 2025). Building upon these advances, recent work has extended reinforcement learning to LLM agents to enhance multi-step decision making (Wang et al., 2025b; Zhai et al., 2024, 2025), long-horizon planning (Song et al., 2024a; Wang et al., 2025a; Xi et al., 2025c), and interaction with external environments (Chen et al., 2025b; Feng et al., 2024; Tan et al., 2024). Across diverse agent settings, RL has been shown to strengthen agents' abilities to decompose complex tasks, coordinate reasoning with action , and adapt policies through environment feedback (Tian et al., 2024), enabling more effective information gathering (Jin et al., 2025; Ramrakhya et al., 2025; Zhang et al., 2025b), iterative self-correction (Kumar et al., 2025; Ma et al., 2025; Zeng et al., 2025b), and tool utilization (Feng et al., 2025; Li et al., 2025a; Zeng et al., 2025a). Despite these successes, most RL-based agent methods are evaluated primarily under in-domain settings with similar task distributions and interfaces, leaving open the question of how well such learned decision-making policies generalize across tasks or transfer to unseen environments.

2.2. Generalization and Forgetting by Post-training

Recent work has examined how post-training strategies affect the generalization and forgetting behaviors of LLMs (Tu et al., 2025; Zhang et al., 2025c; Zhao et al., 2024). Supervised fine-tuning (SFT) is effective for improving in-distribution performance but often leads to over-specialization and degradation of general capabilities due to representation and distributional drift (Kotha et al., 2024; Kumar et al., 2022; Luo et al., 2023; Wang et al., 2024). In contrast, RL, particularly RLVR, tends to better preserve pre-trained representations by optimizing trajectory-level objectives rather than directly fitting target distributions, enabling more robust transfer across tasks and domains (Chen et al., 2025a; Cheng et al., 2025; Chu et al., 2025; Huan et al., 2025). However, RL can also induce negative interference and winner-take-all dynamics, reducing behavioral coverage and leading to reasoning boundary shrinkage (Hu et al., 2025; Nguyen et al., 2025; Sun et al., 2025). Notably, most existing studies focus on static and single-turn LLM tasks, whereas we extend the study of post-training generalization and forgetting to multi-turn LLM agents and systematically evaluate these effects across different environments with distinct observation and action spaces.

3. Preliminaries

3.1. Preliminaries

Task Formulation. We follow the ReAct interaction paradigm (Yao et al., 2022b) to formulate a multi-turn decision-making task. The task can be represented by the tuple $(\mathcal{U}, \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{T}, \mathcal{R})$. Here, $\mathcal{U}$ denotes the instruction space, $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, and $\mathcal{O}$ represents the observation space. The function $\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ denotes the deterministic state transition function, and $\mathcal{R} : \mathcal{U} \times \mathcal{S} \rightarrow \mathbb{R}$ is the reward function.

Given a task instruction $u \in \mathcal{U}$, the agent operates based on a policy π_θ (an LLM parameterized by θ). In each interaction step t, the state s_t consists all previous dialogues and their resulting observations, *i.e.*, $s_t = (a_0, o_0, \ldots, a_{t-1}, o_{t-1})$. The agent generates an action $a_t \sim \pi_\theta(\cdot \mid u, s_t)$. The action a_t includes an internal reasoning trace and a interaction to the environment. The agent then receives an observation $o_k \in \mathcal{O}$ from the environment. This interaction loop continues until the tesk is completed or the maximum number of turns is reached, resulting in a complete trajectory:

$$\tau = \{u, (a_0, o_0), (a_1, o_1), \ldots, (a_T, o_T)\} \tag{1}$$

The environment provides a terminal reward $\mathcal{R}(\tau) \in [0, 1]$ upon task completion.

Policy Gradient. In RL, our objective is to optimize the policy parameters θ to maximize the expected cumulative reward over all possible trajectories for the given task (Sutton and Barto, 1998):

$$\max_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[\mathcal{R}(\tau)] \tag{2}$$

To optimize the objective function $J(\theta)$, we utilize policy gradient methods (Sutton et al., 1999). Unlike value-based methods that approximate a value function, policy gradient methods directly search the policy parameter space. The core idea is to perform gradient ascent to update θ, increasing the probability of trajectories that yield high rewards. The vanilla policy gradient is formulated as:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}\left[\mathcal{R}(\tau) \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t)\right] \tag{3}$$

Based on this gradient estimation, the parameters are updated via $\theta_{new} = \theta_{old} + \alpha \nabla_\theta J(\theta)$, where α is the learning rate. However, standard policy gradient methods often suffer from high variance, leading to unstable training dynamics. To address this challenge, we adopt the widely-used GRPO algorithm (DeepSeek-AI, 2025), detailed in Appendix C.

3.2. Experimental Settings

Environment Setup. We select five representative agent environments, including WebShop (Yao et al., 2022a), SearchQA (Dunn et al., 2017), TextCraft (Sanghi et al., 2022), AlfWorld (Shridhar et al., 2020), and BabyAI (Chevalier-Boisvert et al., 2018), with characteristics of each environment summarized in Table 1. Among them, WebShop is an interactive web shopping environment; SearchQA is a Q&A environment augmented with context from a search engine; ALFWorld requires agents to explore rooms and execute tasks in a household setting; BabyAI is an interactive grid world simulator; and TextCraft is a game environment for crafting items in Minecraft. The task data $\mathcal{U}$ is sourced from AgentGym (Xi et al., 2025b), with detailed statistics and action spaces for each environment provided in Appendix B.

Training Setup. We perform RFT training using the AgentGym-RL framework[1] (Xi et al., 2025c) with Qwen2.5-3B-Instruct and Qwen2.5-7B-Instruct (Team, 2024). Following the ReAct (Yao et al., 2022b) paradigm, each action receives real-time environment feedback. For all experiments, we sample 8 trajectories per task and set the maximum response length to 8192 tokens. Following Xi et al. (2025b), we set the maximum interaction turns K for each environment as follows: 10 for WebShop, 5 for SearchQA, 30 for AlfWorld, 10 for BabyAI, and 15 for TextCraft.

[1] https://github.com/wooodyy/AgentGym-RL

Table 2 | Results of generalization across task difficulties within the same environment.

Models	WebShop			SearchQA			TextCraft			AlfWorld			BabyAI		
	easy	*hard*	*all*	*easy*	*hard*	*all*	*easy*	*hard*	*all*	*easy*	*hard*	*all*	*easy*	*hard*	*all*
							Qwen2.5-3B-Instruct								
base model	21.7	10.6	15.3	63.7	6.5	23.7	23.7	2.3	14.5	26.8	8.0	13.2	71.5	48.0	61.6
train w/ $\mathcal{U}_{\text{easy}}$	90.3	75.3	81.6	82.7	16.9	36.6	97.6	49.4	76.9	93.2	82.0	85.1	93.4	79.0	87.3
train w/ $\mathcal{U}_{\text{hard}}$	86.1	84.5	85.2	72.5	19.4	35.3	95.0	42.2	72.3	96.1	92.9	93.8	92.3	77.3	86.0
train w/ $\mathcal{U}$	92.8	84.3	87.9	87.6	22.1	41.8	93.9	47.4	73.9	97.0	89.8	91.8	93.2	77.5	86.6
$\mathcal{U}_{\text{easy}} + \mathcal{U}_{\text{hard}}$	93.6	85.2	88.7	82.1	19.8	38.5	93.9	52.9	76.3	97.3	93.4	94.4	94.5	82.2	89.3
$\mathcal{U}_{\text{hard}} + \mathcal{U}_{\text{easy}}$	90.2	82.4	85.7	82.0	18.7	37.7	97.8	40.1	73.0	95.5	92.3	93.6	94.5	80.4	88.6
							Qwen2.5-7B-Instruct								
base model	44.1	17.4	28.6	79.6	10.4	31.2	46.9	16.0	33.6	40.2	21.4	26.6	80.1	47.9	67.0
train w/ $\mathcal{U}_{\text{easy}}$	88.1	77.5	82.0	85.8	21.1	40.5	98.2	47.4	76.4	95.2	89.7	91.2	95.2	76.1	87.2
train w/ $\mathcal{U}_{\text{hard}}$	87.9	81.4	84.2	93.3	27.0	46.9	99.3	57.3	81.3	97.5	94.6	95.4	92.8	77.9	86.5
train w/ $\mathcal{U}$	92.9	81.9	86.5	91.8	26.6	46.1	95.2	52.3	80.9	97.7	97.2	92.0	95.6	74.5	88.8
$\mathcal{U}_{\text{easy}} + \mathcal{U}_{\text{hard}}$	90.6	77.5	83.0	89.8	25.5	44.8	98.7	64.0	83.8	96.1	92.6	93.6	95.5	82.9	90.2
$\mathcal{U}_{\text{hard}} + \mathcal{U}_{\text{easy}}$	89.1	79.1	83.3	87.6	21.5	42.3	99.8	57.0	81.4	97.7	88.5	91.1	92.5	78.2	86.5

Evaluation Setup. As summarized in Table 1, WebShop and BabyAI provide dense rewards, while SearchQA, AlfWorld, and TextCraft use binary rewards. We follow DeepSeek-AI (2025) and adopt exact matching, counting an action as correct only if it exactly matches the reference. To reduce randomness, we report avg@8 as the main metric, and also measure the average interaction turns ($\bar{k}$) and generated tokens ($\bar{l}$) for efficiency. For fair comparison, all agents are evaluated with the maximum number of interaction turns set to $K = 20$ during testing.

4. Does RFT Yield Generalization Across Task Difficulties within the Same Environment?

Reinforcement fine-tuning trains LLM agents through interaction with the environment, allowing them to learn its dynamics and adapt over time (Cui et al., 2025; Song et al., 2024b). Yet even with a fixed action and observation space, tasks within the same environment can differ substantially in exploration depth and information accessibility. In this section, we investigate *whether RFT policies learned on a subset of tasks transfer to other tasks of differing difficulty within the same environment*.

Setting. Following practice in previous work (Bengio et al., 2009; Mukherjee et al., 2023), we categorize the tasks $\mathcal{U}$ into *easy* and *hard* difficulty levels (denoted as $\mathcal{U}_{\text{easy}}$ and $\mathcal{U}_{\text{hard}}$) based on the avg@8 results of the Qwen2.5-7B-Instruct model, while ensuring a balanced distribution of data between the two difficulty levels. The same categorization is applied to the test set. Detailed data statistics for each environment are provided in Appendix B.

RFT demonstrates strong tranferability across varying difficulty levels within the same environment. As shown in Table 2, RFT demonstrates strong robustness when trained on data of varying difficulty levels within the same environment. Notably, with 7B model on WebShop, training on $\mathcal{U}_{\text{easy}}$ improves performance on the *hard* testset by 60.1 points, suggesting that RFT encourages agents to adapt to the environment, thereby enabling performance transfer across tasks with different difficulty levels and varying steps (Chen et al., 2025a; Huan et al., 2025). Overall, we also find that training on $\mathcal{U}_{\text{hard}}$ yields more gains on test set, e.g., for 3B model in AlfWorld, training on $\mathcal{U}_{\text{hard}}$ outperforms training on $\mathcal{U}_{\text{easy}}$ by 8.7 points. We attribute this trend to the richer failure signals and longer-horizon exploration induced by harder tasks, which push the RFT optimizer to seek better policies (Wang et al., 2023; Xu et al., 2023).

Curriculum learning can further enhance performance. Additionally, we investigate the impact of the training sequence using data of varying difficulty levels. Results in Table 2 indicate that training on $\mathcal{U}$ (i.e., mixture of $\mathcal{U}_{\text{easy}}$ and $\mathcal{U}_{\text{hard}}$) generally does not yield optimal performance. In most cases, training on $\mathcal{U}_{\text{easy}}$ first, followed by $\mathcal{U}_{\text{hard}}$, achieves the best results. Compared with using a single difficulty level, this easy-to-hard curriculum enables further performance improvement. For example, on BabyAI, training on $\mathcal{U}_{\text{easy}} + \mathcal{U}_{\text{hard}}$ outperforms training on $\mathcal{U}_{\text{easy}}$ and on $\mathcal{U}_{\text{hard}}$ by 2.0 and 3.3 points, respectively. This result validates the rationale and effectiveness of curriculum learning (Qi et al., 2024; Zhang et al., 2025a).

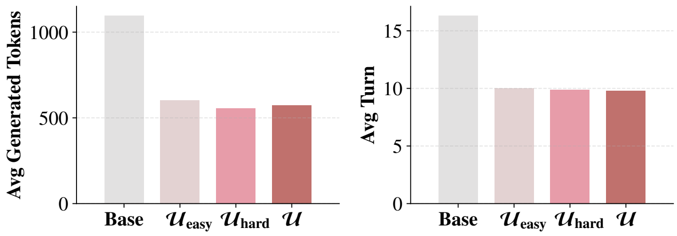


Figure 2 | Average generated tokens and interactive turn of all environments for Qwen2.5-3B-Instruct model trained with varying-difficulty tasks.

RFT training encourages agents to perform efficient exploration. Figure 2 reports the average number of interaction turns and generated tokens across different environments, with a more detailed breakdown provided in Figure 8 of Appendix E. We find that RFT enables agents to interact and explore more efficiently within the same environment. For example, when training the 3B model on $\mathcal{U}_{\text{easy}}$ in BabyAI, the average interaction turns decrease from 10.76 to 4.19, and the average trajectory length is reduced from 624.58 to 160.60 tokens. These results indicate that RFT not only improves success rates but also encourages more concise and goal-directed exploration, substantially enhancing interaction efficiency (Nakano et al., 2021; Song et al., 2024b). A case can be found in Appendix G.

5. Does RFT Yield Generalization Across Different Environments?

In real-world scenarios, agents may encounter previously unseen environments and tasks, which is essential for practical deployment. We therefore investigate *whether RFT improves performance in unseen environments with different background knowledge, observation space and action spaces*. Concretely, we perform RFT training within a single environment and then evaluate the trained agent on other environments to measure cross-environment generalization.

Setting. We employ the base model (i.e., Qwen2.5-3B-Instruct or Qwen2.5-7B-Instruct) as the baseline and evaluate its `avg@8` metric across varying environments. To make these comparisons explicit, we define three metrics: ΔHeld-In, ΔHeld-Out, and ΔOverall. ΔHeld-In measures the average improvement over the baseline when the training and test environments coincide. ΔHeld-Out measures the average improvement when the agent is evaluated on environments different from those seen during training. Finally, ΔOverall reports the average improvement over the baseline across all evaluation settings.

Performance differs significantly between held-in and held-out environments. Table 3 reveals that agents exhibit generalization across different environments, yet a significant performance gap exists between held-in and held-out settings. Specifically, substantial gains are achieved under held-

Table 3 | Results of generalization across different environments. Darker red indicates greater performance improvement, while darker blue signifies more pronounced performance decline. The best result is in **bold**, while the second-best is marked with <u>underline</u>.

Models	WebShop	SearchQA	TextCraft	AlfWorld	BabyAI	ΔHeld-In	ΔHeld-Out	ΔOverall
				Qwen2.5-3B-Instruct				
base model	15.30	23.66	14.50	13.19	61.83	–	–	–
train w/ WebShop	87.86	25.84	18.50	23.06	70.91	<u>+72.56</u>	**+6.29**	<u>+19.54</u>
train w/ SearchQA	22.97	41.78	22.75	12.13	65.72	+18.13	+4.69	+ 4.56
train w/ TextCraft	14.46	22.16	73.88	11.00	63.73	+59.38	−0.66	+11.35
train w/ AlfWorld	21.86	23.50	24.88	91.81	64.68	**+78.62**	<u>+4.91</u>	**+19.65**
train w/ BabyAI	28.36	22.63	16.75	4.50	86.55	+24.72	+1.40	+ 6.06
				Qwen2.5-7B-Instruct				
base model	28.59	31.19	33.63	26.56	67.00	–	–	–
train w/ WebShop	86.50	33.28	40.75	24.13	79.21	<u>+57.91</u>	+4.75	<u>+15.38</u>
train w/ SearchQA	47.07	46.12	35.25	16.75	80.33	+14.93	<u>+5.91</u>	+ 7.71
train w/ TextCraft	38.30	32.19	80.88	31.50	77.95	+47.25	**+6.65**	+14.77
train w/ AlfWorld	34.31	29.59	36.13	92.00	72.91	**+65.44**	+3.13	**+15.59**
train w/ BabyAI	10.25	29.41	39.25	28.13	88.79	+21.79	−3.23	+ 1.77

in conditions. In AlfWorld, performance improves by 78.62 and 65.44 points for the 3B and 7B models, respectively. In contrast, in held-out conditions, generalization remains possible in most cases, but with more modest gains. On average, the 3B and 7B models yield improvements of 3.32 and 3.44 points on unseen environments, respectively.

In unseen environments, positive transfer is observed in most cases. Compared to the baseline, most trained agents can demonstrate transferability to unseen environments, even when these environments demand different background knowledge and operate under different action spaces. Notably, agents trained on WebShop, AlfWorld, and SearchQA consistently exhibit positive Δheld-out. When evaluated on WebShop, agents trained on SearchQA yield performance gains of 7.67 and 18.48 points for the 3B and 7B models, respectively. This can be attributed to the similarity between WebShop and SearchQA as search-based environments. Specifically, the agent trained on SearchQA learns to formulate more flexible search queries, as well as efficient information extraction from complex results. An illustrative case is presented in Figure 10 in Appendix G and discussed in Section 7.

In unseen environments, negative effects may sometimes occur. However, agents trained on TextCraft and BabyAI may struggle to generalize to other environments, e.g., after training on BabyAI, the 7B models show average negative improvements of −3.23 on held-out tasks, and even drop sharply from 28.59 to 10.25 on WebShop. Through careful analysis, we find that because the BabyAI environment provides available actions at each step, the agent gradually becomes dependent on this information during training, leading to a decline in long-horizon reasoning capability. When faced with other environments, it fails to accurately use the valid action, resulting in a sharp performance drop. A case of interaction between the agent and BabyAI environment is provided in Appendix G.

Cross-environment generalization performance varies significantly across target environments. Moreover, our observations reveal that cross-environment generalization performance is highly dependent on the target environment. On the one hand, agents trained on nearly all source

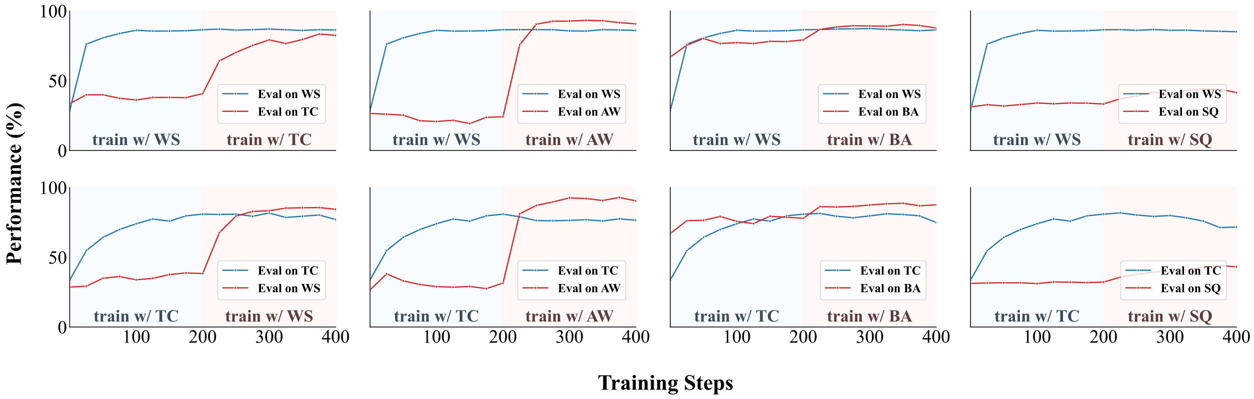


Figure 3 | Training dynamics of forgetting and transfer in sequential two-stage cross-environment training with Qwen2.5-7B-Instruct, where blue and red denote the upstream environment and the downstream environment, respectively.

environments generalize effectively to TextCraft and BabyAI, as these domains rely less on specific background knowledge, thereby allowing acquired skills to transfer more readily. On the other hand, effective transfer to AlfWorld and SearchQA proves to be significantly more challenging. We attribute this to the strict action validation and sparse feedback inherent in these two environments. For instance, AlfWorld responds uniformly with "Nothing happens." to all invalid actions, offering no instructional guidance for improvement.

6. How Does Sequential RFT Across Environments Affect Transfer and Forgetting?

Beyond zero-shot transfer to unseen environments without additional training, we further investigate the effect of sequential training across multiple environments on agentic behaviors, specifically the resulting learning dynamics on memorizing downstream tasks or forgetting on upstream ones. In our experimental setup, a model initially converged in one environment is further trained on a second environment, after which we assess both its retention of performance on the original environment and its adaptation to the new one. Finally, we extend this analysis to sequential training across five environments, comparing this strategy with joint training across a mixture of environments.

Setting. We conduct 20 two-stage experiments by sequentially pairing each of the five environments as the upstream and the downstream, as well as several five-stage training experiments. Using agents trained on single-environment as baselines, we evaluate anti-forgetting by upstream performance retention and transferability by downstream performance gains. Furthermore, we compare the five-stage sequential approach with joint training, in which the agent is trained on all available data from every environment in a randomly mixed manner.

Sequential training demonstrates consistent anti-forgetting and transferability. Figure 3 illustrates the performance dynamics of 8 two-stage sequential training scenarios, with others and the final results summarized in Figure 7 and Table 6 in Appendix D. Overall, sequential training matches or exceeds single-task performance on the downstream environment while largely preserving upstream performance. For instance, when the WebShop-pre-trained agent is further trained on TextCraft, it boosts performance on TextCraft from the single-task baseline of 80.88 to 82.50. Meanwhile, performance on WebShop experiences only a minor fluctuation, shifting from 86.5 to 86.32.

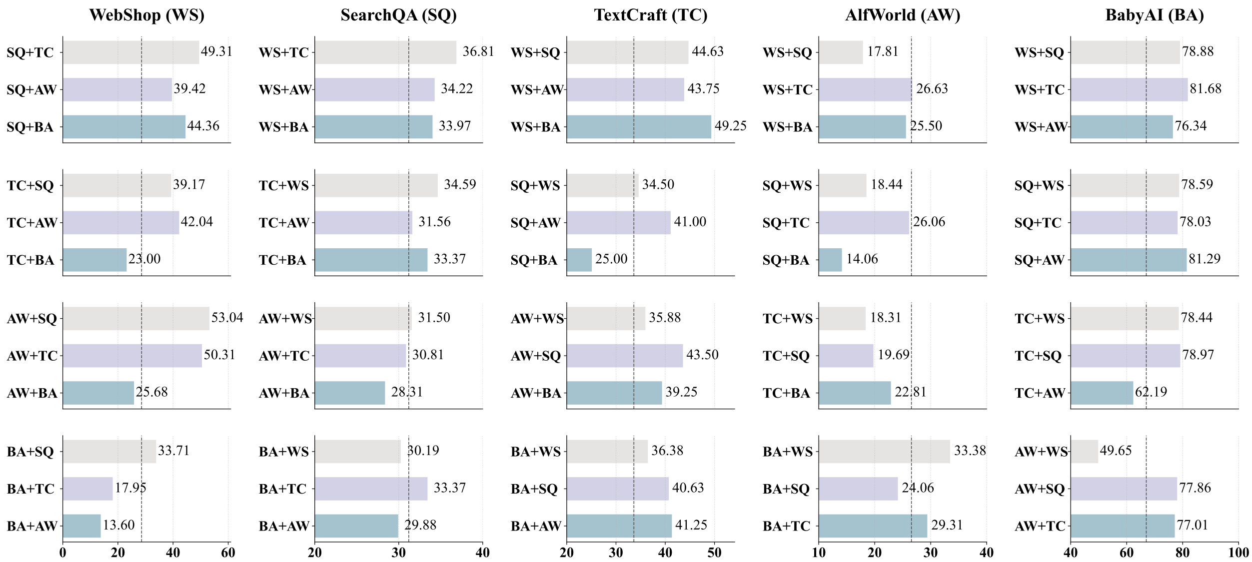


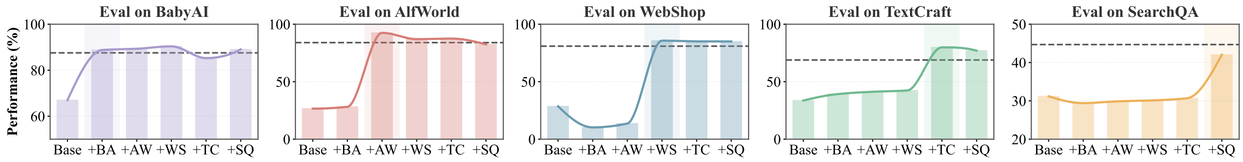
Figure 4 | Generalization of sequential two-stage cross-environment training with Qwen2.5-7B-Instruct. Across five environments (WS, SQ, TC, AW, BA), the figure presents the generalization performance on the *three unseen environments* following sequential training on two environments. Each subplot corresponds to a fixed first environment, with *dashed line* indicating the baseline performance.

These findings suggest that sequential training endows agents with stable capabilities for transfer and resistance to forgetting.

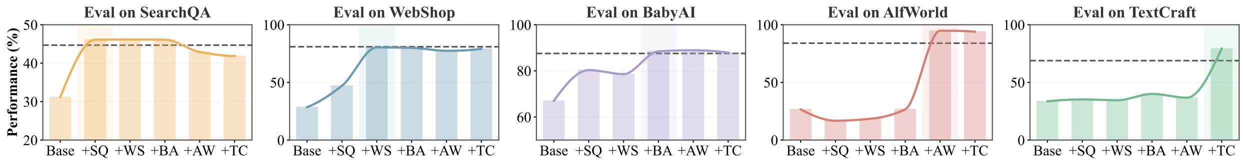
Generalization in multi-environment training is highly correlated with single-environment training. Figure 4 demonstrates the generalization performance of multi-environment training, revealing a strong alignment with the generalization patterns in single-environment training. First, environments that yield poor generalization in single-environment settings continue to be detrimental. For instance, the agent trained on BabyAI exhibits severe negative effect on WebShop. Even when BabyAI is trained sequentially after other generalizable environments (i.e., TC+BA, AW+BA), it drastically degrades their WebShop performance. Notably, the score of TC+BA dropped sharply from 38.30 (score of TC in Table 3) to 23.0.

Moreover, from the perspective of target environments, the results also show consistency with single-environment generalization. For target environments easy to generalization in single-environment scenarios, such as TextCraft and BabyAI, sequentially trained agents also tend to perform well. Specifically, agents pre-trained on WebShop (i.e., WS+SQ, WS+AW, WS+BA) achieve significant gains on TextCraft, increasing by 11.00, 10.12, and 15.62 points, respectively. However, for environments challenging for generalization, like Alf World, sequential training yields limited benefits. For instance, all three agents utilizing TextCraft as the upstream environment (TC+WS, TC+SQ, TC+BA) suffer performance degradation on Alf World.

Training order significantly affects generalization performance in held-out environments. As shown in Figure 4, the training order exerts a substantial influence on generalization performance. For instance, when evaluated on TextCraft and Alf World, BA+SQ outperforms SQ+BA by 15.63 and 10.00 points, respectively. This represents a substantial margin, particularly for held-out scenarios. We attribute this phenomenon to the inherent task difficulty within each environment. As noted in Section 5, BabyAI provides detailed feedback, whereas SearchQA imposes strict constrains with



(a) Sequential training of BabyAI, AlfWorld, WebShop, TextCraft, and SearchQA.



(b) Sequential training of SearchQA, WebShop, BabyAI, AlfWorld, and TextCraft.

Figure 5 | Training dynamics of sequential training across five environments. We present the results for representative sequence combinations, monitoring how performance on each environment changes as the agent is trained on different environments sequentially. The *dashed lines* denote the performance achieved by **joint training on a mixture of data from all five environments**.

limited feedback. Consequently, the BA+SQ order naturally creates an "easy-to-hard" curriculum, which in turn facilitates better generalization performance.

Sequential training achieves performance comparable to joint training. Furthermore, we conduct sequential training over five environments in different orders. Figure 5 illustrates the performance dynamics on each environment across the five training stages for two representative sequences; for comparison, joint-training results are indicated by dashed lines. The results show that sequential training achieves performance comparable to joint training, even after training on five distinct tasks. Overall, the final performance is insensitive to the training order, which we attribute to RFT's ability to preserve previously acquired capabilities, consistent with findings in prior work. For environments such as AlfWorld and SearchQA—to which other tasks struggle to generalize—relatively pronounced forgetting may occur over the course of long-term sequential training.

7. Further Analysis and Discussion

Failure Mode Analysis. We conduct a fine-grained error analysis on both intra/inter-environment evaluations, summarizing 8 common failure modes including *instruction misinterpretation*, *action execution failure*, and *logical deficits*. Using GPT-5-mini, we systematically categorize the error trajectories across all scenarios. More detail about each failure pattern can be found in Appendix F.

The results presented in Figure 6 reveal that while failure modes differ by environment, errors related to "Confirmation Bias" are prevalent (> 10%) across all scenarios. This suggests that after training, agents tend to exhibit overconfidence, neglecting further verification and lacking the capacity for self-reflection based on environmental feedback. Moreover, in SearchQA, errors categorized as "Guessing or Fabrication" are widespread among both held-in (23.8%) and held-out (21.2%) settings. This highlights a critical deficiency in tool utilization, which fundamentally constrains the agent's generalization potential.

When the training and test environments differ, agents exhibit different failure modes. For example, in WebShop we observe an increase in the proportion of "State or Memory Inconsistency" errors, rising from 4.3% in held-in to 21.9% in held-out settings. This suggests that, when confronted with large volumes of information, the agent's decision-making becomes less coherent, and it struggles to

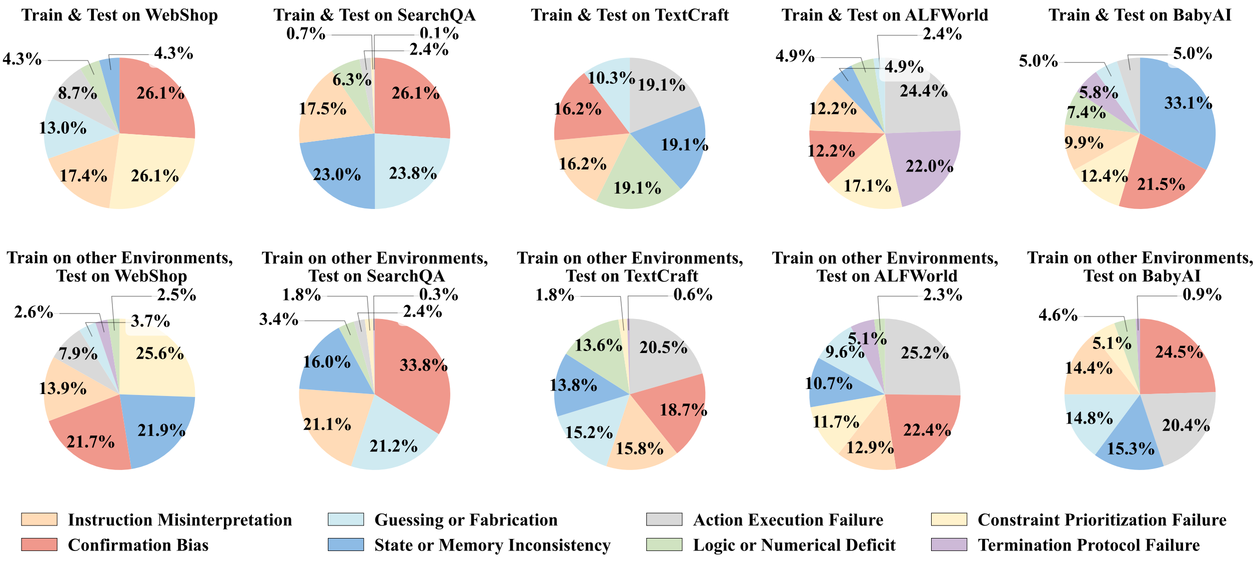


Figure 6 | Comparison of failure mode distributions between in-environment and out-of-environment evaluation.

generalize its ability to extract salient information. A full environment-by-environment breakdown is provided in Figure 9.

Case Study. We perform a comprehensive analysis of environment generalization and present specific case studies in Appendix G. Notably, Figure 10 illustrates the mechanism behind the successful transfer from SearchQA to WebShop. Specifically, the base agent tends to blindly input the entire instruction into the search bar, and struggles to accurately extract key information from the voluminous HTML content, resulting in the retrieval and selection of irrelevant items. In contrast, the agent trained on SearchQA learns to effectively search for key details and extract information, successfully selecting the correct item.

Furthermore, Figure 11 offers qualitative insight into why SearchQA generalization remains difficult, using an AlfWorld-trained agent as a case study. Although both agents fail initially, the SearchQA-trained agent iteratively refines its queries to become more specific, improving retrieval and ultimately recovering. By contrast, this query-refinement behavior does not reliably transfer: the AlfWorld-trained agent falls into a degenerate loop, repeatedly issuing near-duplicate searches, returning the same responses, and ultimately failing.

8. Conclusion

In this paper, we present a systematic study of how RFT affects the transfer and generalization of LLM agents for multi-turn decision-making. Through large-scale experiments along three complementary axes, we characterize when RFT generalizes within and across environments and identify generalization patterns. We further complement our quantitative results with failure mode analysis and qualitative case study to pinpoint where agents break down and what behaviors fail to transfer. Together, our findings offer practical guidance for training and evaluating agents under distribution shift, and we hope they inform the development of agents that generalize reliably in real-world deployments.

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Appendix

A. Limitations and Future Work

This paper adopts a new perspective to investigate how reinforcement fine-tuning affects the generalization ability of LLM agents. Our experiments focus on the Qwen2.5 family and yield extensive empirical results together with substantive insights. Despite this progress, important challenges remain. First, we rely on default training/evaluation protocols and hyperparameter choices, without exhaustive tuning; more careful optimization may bring new findings. Second, we adopt GRPO—the most commonly used RL algorithm for LLM agents at present—rather than other approaches such as REINFORCE++ (Hu, 2025) or DAPO (Yu et al., 2025); we leave these for future work. Additionally, due to computational constraints, our sequential multi-environment training study does not enumerate all possible environment orderings; instead, we evaluate a small set of representative sequence. Although these experiments already reveal important insights, a broader and more fine-grained investigation of ordering effects is left for future work and may further substantiate our findings.

B. Dataset Details

We select five representative agent environments, including WebShop (Yao et al., 2022a), SearchQA (Dunn et al., 2017), TextCraft (Sanghi et al., 2022), AlfWorld (Shridhar et al., 2020), and BabyAI (Chevalier-Boisvert et al., 2018). The characteristics of each environment are shown in Table 1 in Section 3.2. Here, we provide their detailed types and action spaces in Table 4.

Table 4 | Detailed action spaces for each environment.

Environment	Types	Action Spaces
WebShop	*Web Navigation*	*search, click*
SearchQA	*Q&A Search*	*search, answer*
TextCraft	*Text-based Game*	*get, craft, inventory*
AlfWorld	*Household*	*go to, open, close, take from, put in/on, use, heat, cool, clean, slice, inventory, look, examine*
BabyAI	*Embodied*	*turn right, turn left, move forward, go to, go through, toggle and go through, toggle, pickup, drop, check available actions*

Our data is sourced from AgentGym (Xi et al., 2025b). Following practice in previous work (Bengio et al., 2009; Mukherjee et al., 2023), we categorize the tasks $\mathcal{U}$ into *easy* and *hard* difficulty levels (denoted as $\mathcal{U}_{\text{easy}}$ and $\mathcal{U}_{\text{hard}}$) based on the avg@8 results of the Qwen2.5-7B-Instruct model, while ensuring a balanced distribution of data between the two difficulty levels. The same categorization is applied to the test set. Detailed statistics for each environment are provided in Table 5.

Table 5 | Detailed data statistics for each environment.

Environment	Training dataset			Testing dataset		
	easy	*hard*	*all*	*easy*	*hard*	*all*
WebShop	2104	1826	3930	84	116	200
SearchQA	1960	2040	4000	120	280	400
TextCraft	235	209	444	57	43	100
AlfWorld	1267	1153	2420	55	145	200
BabyAI	398	412	810	52	38	90

C. Detailed algorithm of GRPO.

Group Relative Policy Optimization (GRPO) is an efficient online reinforcement learning algorithm tailored for LLMs. It eliminates the need for a separate critic network typically required in PPO, thereby reducing computational overhead. Instead, GRPO estimates the baseline using *group relative advantages* to significantly reduce gradient variance.

Specifically, for each input query q (derived from u), GRPO samples a group of outputs $\{y_1, y_2, \ldots, y_G\}$ from the old policy $\pi_{\theta_{old}}$ and obtains their corresponding rewards $\{R_1, R_2, \ldots, R_G\}$. The advantage A_i for each output is calculated by normalizing the rewards within the group:

$$A_i = \frac{r_i - \mathrm{mean}(\{R_1, \ldots, R_G\})}{\mathrm{std}(\{R_1, \ldots, R_G\})} \tag{4}$$

Finally, GRPO updates the policy by maximizing the following surrogate objective, which incorporates PPO-style clipping (Schulman et al., 2017) and a KL divergence penalty to ensure training stability:

$$\mathcal{J}_{\mathrm{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\Big(\min\big(r_i(\theta)A_i,\ \mathrm{clip}\big(r_i(\theta), 1-\epsilon, 1+\epsilon\big)A_i\big) - \beta\mathbb{D}_{\mathrm{KL}}\Big)\right], \tag{5}$$

where $r_i(\theta) = \frac{\pi_\theta(y_i|q)}{\pi_{\theta_{old}}(y_i|q)}$ denotes the probability ratio between the new and old policies, $\{y_i\}_{i=1}^{G} \sim \pi_{\theta_{\mathrm{old}}}(q)$ represents the outputs y_i sampled from the old policy $\pi_{\theta_{\mathrm{old}}}$ given the query q, ϵ is the clipping parameter, and β is the coefficient for the KL divergence term.

D. Detailed Results of Sequential Cross-Environment Training

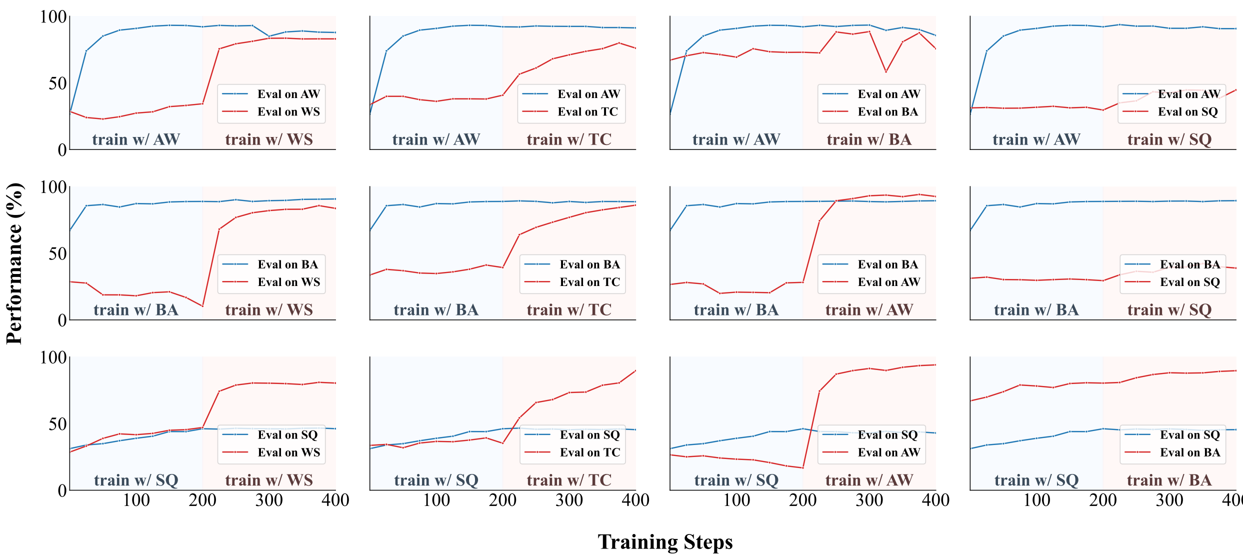

Figure 7 | Training dynamics of forgetting and transfer in other sequential two-stage cross-environment training with Qwen2.5-7B-Instruct, where blue and red denote the upstream environment and the downstream environment, respectively.

Figure 3 in Section 6 illustrates the training dynamics for 8 two-stage sequential training configurations. Here, we present the dynamics for the remaining 12 pairs in Figure 7. Additionally, detailed final results are reported in Table 6.

Table 6 | Results of sequential training across different environments with Qwen2.5-7B-Instruct model.

Upstream	Downstream	WebShop	SearchQA	TextCraft	ALFWorld	BabyAI
base model		28.59	31.19	33.63	26.56	67.00
WebShop		86.50	33.28	40.75	24.13	79.21
	+ SearchQA	85.08	41.44	44.63	17.81	78.88
	+ TextCraft	86.32	36.81	82.50	26.63	81.68
	+ ALFWorld	85.99	34.22	43.75	90.69	76.34
	+ BabyAI	86.38	33.97	49.25	25.50	87.74
SearchQA		47.07	46.12	35.25	16.75	80.33
	+ WebShop	80.35	46.16	34.50	18.44	78.59
	+ TextCraft	49.31	46.19	78.63	26.06	78.03
	+ ALFWorld	39.42	42.92	41.00	94.00	81.29
	+ BabyAI	44.36	45.44	25.00	14.06	89.62
TextCraft		38.30	32.19	80.88	31.50	77.95
	+ WebShop	84.33	34.59	77.00	18.31	78.44
	+ SearchQA	39.17	43.16	71.63	19.69	78.97
	+ ALFWorld	42.04	31.56	76.50	90.38	62.19
	+ BabyAI	23.00	33.37	74.88	22.81	87.55
ALFWorld		34.31	29.59	36.13	92.00	72.91
	+ WebShop	82.95	31.50	35.88	87.75	49.65
	+ SearchQA	53.04	44.66	43.50	90.63	77.86
	+ TextCraft	50.31	30.81	76.00	91.19	77.01
	+ BabyAI	25.68	28.31	39.25	85.56	75.59
BabyAI		10.25	29.41	39.25	28.13	88.79
	+ WebShop	83.60	30.19	36.38	33.38	90.66
	+ SearchQA	33.71	38.84	40.63	24.06	89.37
	+ TextCraft	17.95	33.37	86.00	29.31	88.60
	+ ALFWorld	13.60	29.88	41.25	92.50	89.32

E. Detailed Results of average turns and generated tokens in different environment

Figure 2 in section 4 reports the average number of interaction turns and generated tokens across different environments. The more detailed results are reported in Figure 8.

F. Failure Mode Analysis

Through detailed analysis, we summarize 8 common error types:

1. **Instruction Misinterpretation**: The agent fails to understand the prompt correctly, does not follow the instructions as indicated, or generates the wrong answer format.
2. **Guessing or Fabrication**: The agent relies too heavily on internal parameters or makes unsupported guesses, instead of utilizing environmental tools to gather necessary external information.
3. **Action Execution Failure**: The agent generates actions that cannot be interpreted by the environment or do not correspond to specific objects within the environment.

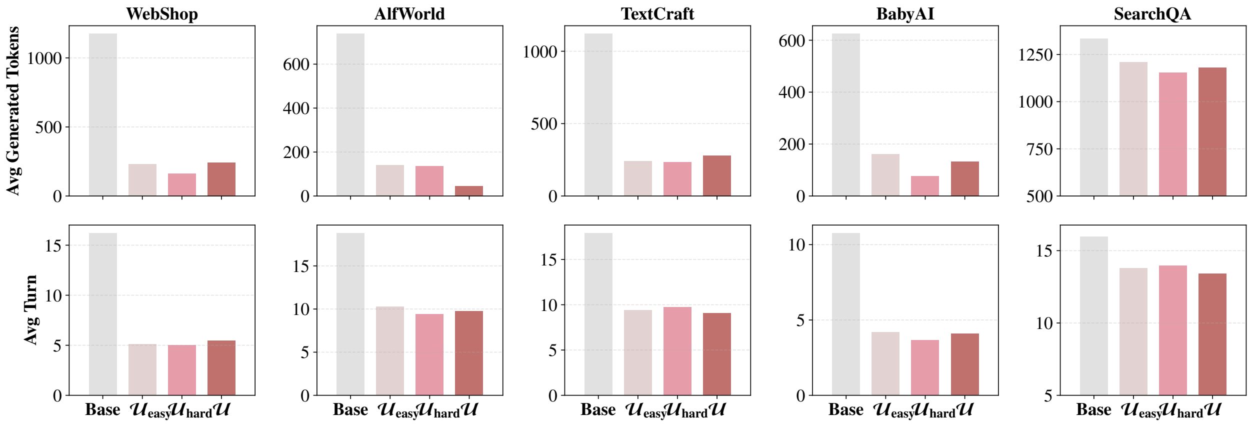

Figure 8 | Average generated tokens and average turn across different environments for Qwen2.5-3B-Instruct model trained with varying difficulties.

4. **Constraint Prioritization Failure**: The agent recognizes multiple constraints in the instructions but fails to evaluate their relative importance correctly, resulting in the violation of core constraints in favor of secondary ones.
5. **Confirmation Bias**: The agent becomes confident that it has found the correct answer or completed key steps, but does not proceed with further verification. Alternatively, it may forcefully apply other clues or validate erroneous information from the prompt.
6. **State or Memory Inconsistency**: The agent exhibits contradictions over time, forgetting tools it has recently invoked or results it has obtained. Completed steps may be repeated unnecessarily.
7. **Logic or Numerical Deficit**: The agent makes errors in logical or numerical reasoning.
8. **Termination Protocol Failure**: The agent fails to issue the environment-specific termination command or prematurely terminates before completing the task.

The results of both in-domain and out-of-domain evaluations are discussed in Section 7. Here, we present a detailed error classification in Figure 9.

G. Case Study

In this section, we present specific case studies. It is worth noting that while all interactions are inherently text-based, we provide visualizations for selected cases to enhance clarity. Additionally, due to the excessive number of interaction turns, we omit the majority of intermediate steps, highlighting only the pivotal moments in the figures.

Figure 10 illustrates a case where a model trained on SearchQA is evaluated on WebShop, comparing its generated trajectory against that of the base model on the same task. As observed in the case, the base model tends to blindly input the entire content of the instruction into the search bar, resulting in the retrieval of numerous irrelevant items. Furthermore, the base model exhibits incoherent decision-making under multiple constraints and struggles to accurately extract key information from the voluminous HTML content returned by the environment, leading to the selection of items from incorrect categories. In contrast, the model trained on SearchQA learns to formulate more flexible search queries, as well as perform efficient information extraction from complex results, thereby enabling it to successfully retrieve key information and select the correct item.

Figure 11 highlights a key reason why agents struggle to generalize to SearchQA, using model

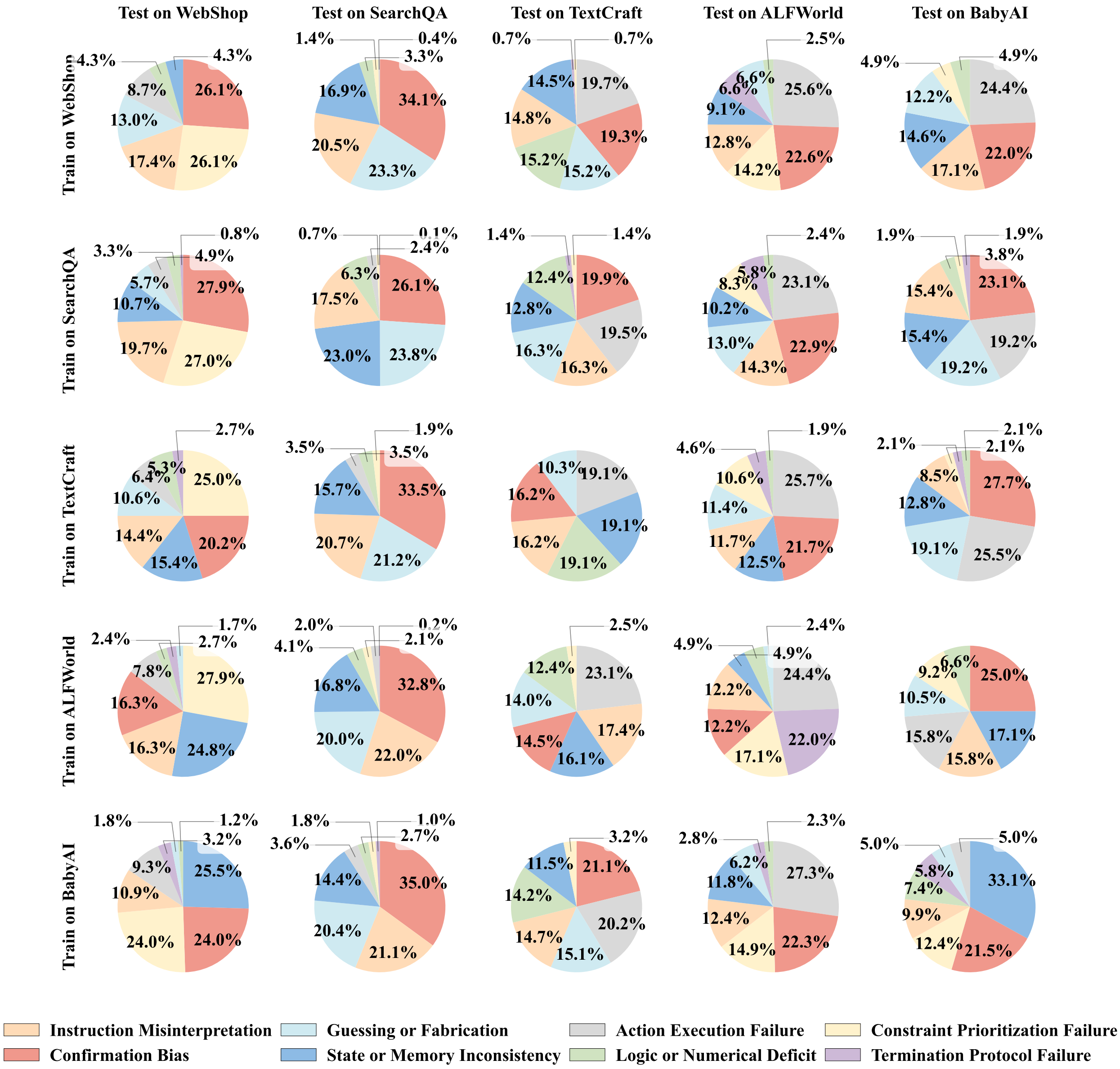

Figure 9 | Failure mode distribution across all train-test environment combinations.

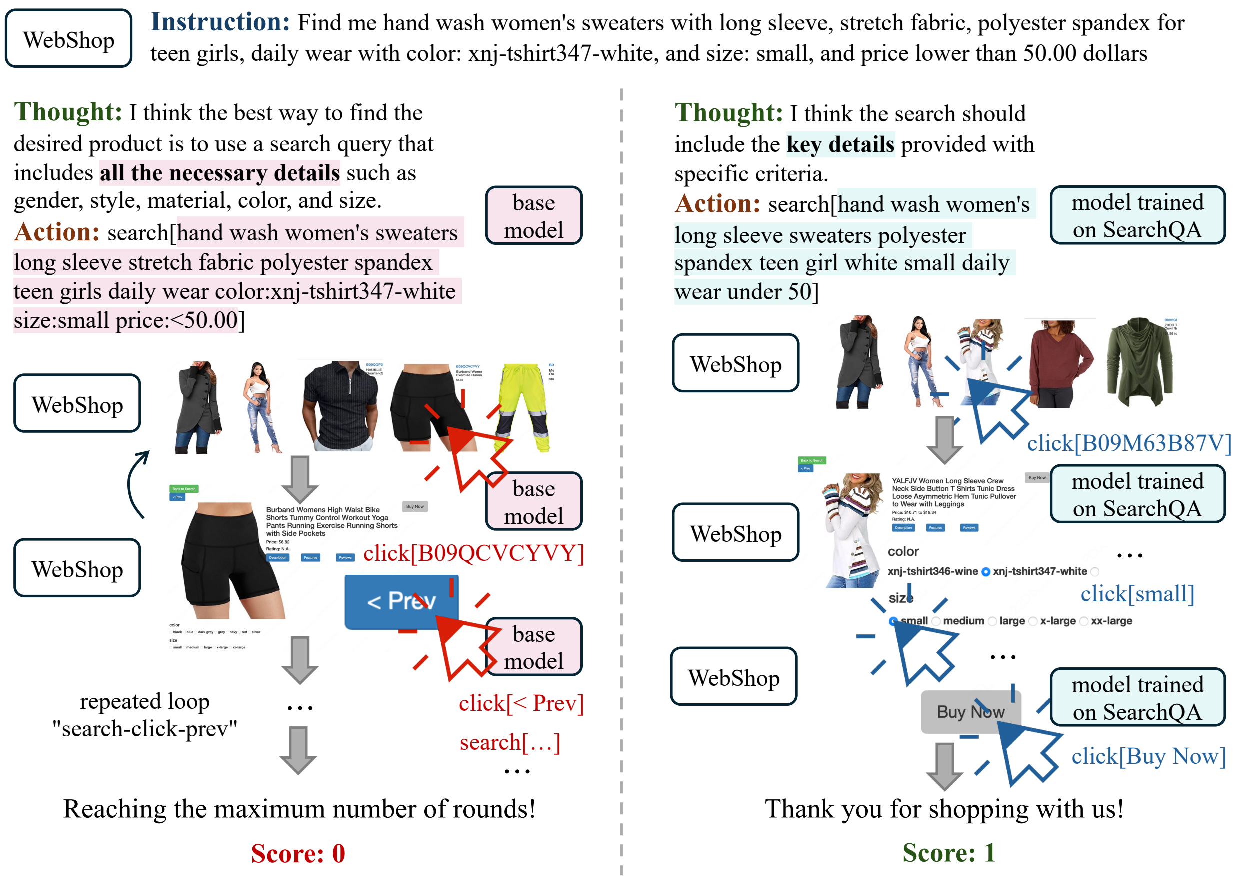

Figure 10 | Comparison of trajectories on WebShop between the base model and the model trained on SearchQA.

trained on AlfWorld as a case study. Although both models fail in their initial attempts, the model trained on SearchQA demonstrates the ability to refine its search queries to achieve greater precision. This capability proves difficult to transfer from other environments, e.g., the model trained on AlfWorld falls into a repetitive loop of the same searches and answers, consequently failing to resolve the task.

Figure 12 illustrates how training enhances exploration efficiency, using BabyAI as a case study. The base model, lacking strong spatial awareness, struggles to accurately pinpoint the target location based on textual descriptions. Consequently, it falls into a pattern of redundant exploration; although it eventually completes the task, it requires an excessive number of interaction turns. In contrast, the trained model is able to precisely locate the target, thereby completing the task via an optimal path and significantly improving exploration efficiency.

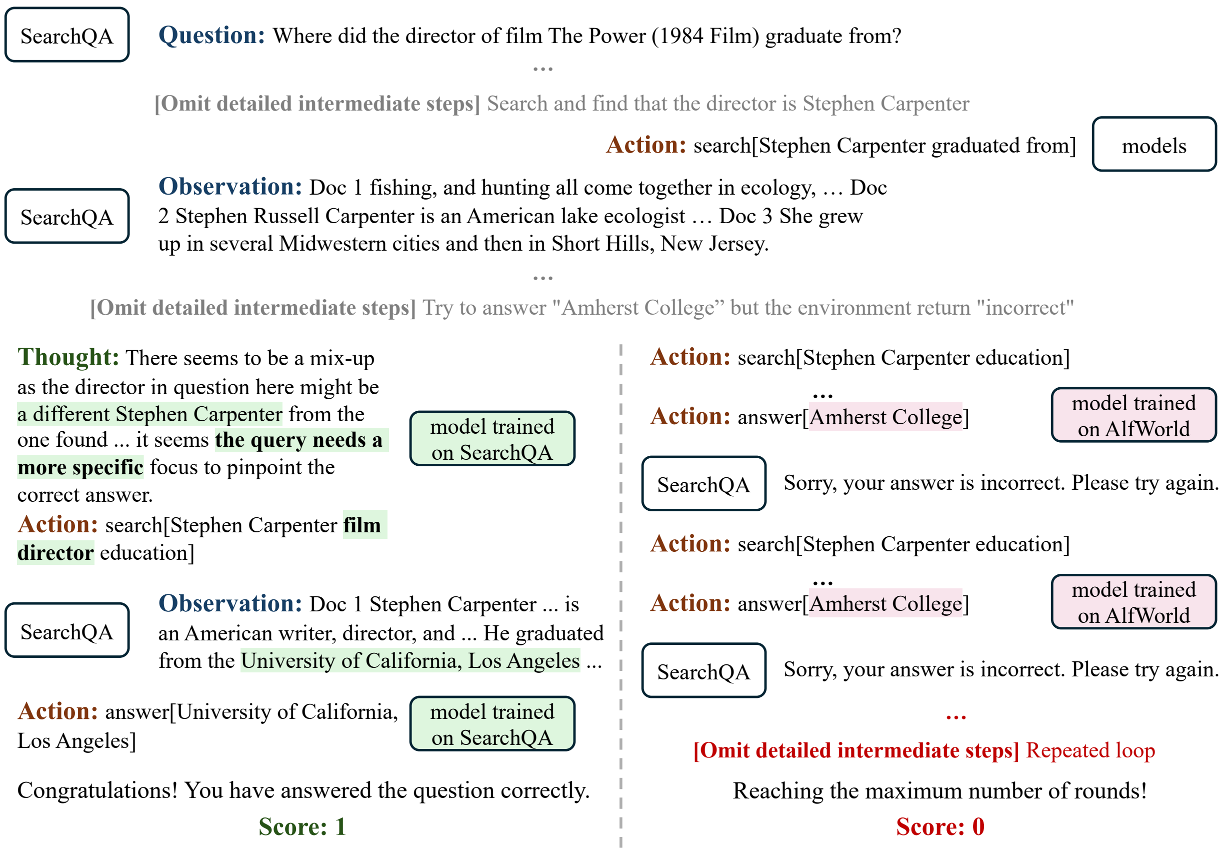

Figure 11 | Comparison of trajectories on SearchQA between the model trained on SearchQA and the mode trained on AlfWorld.

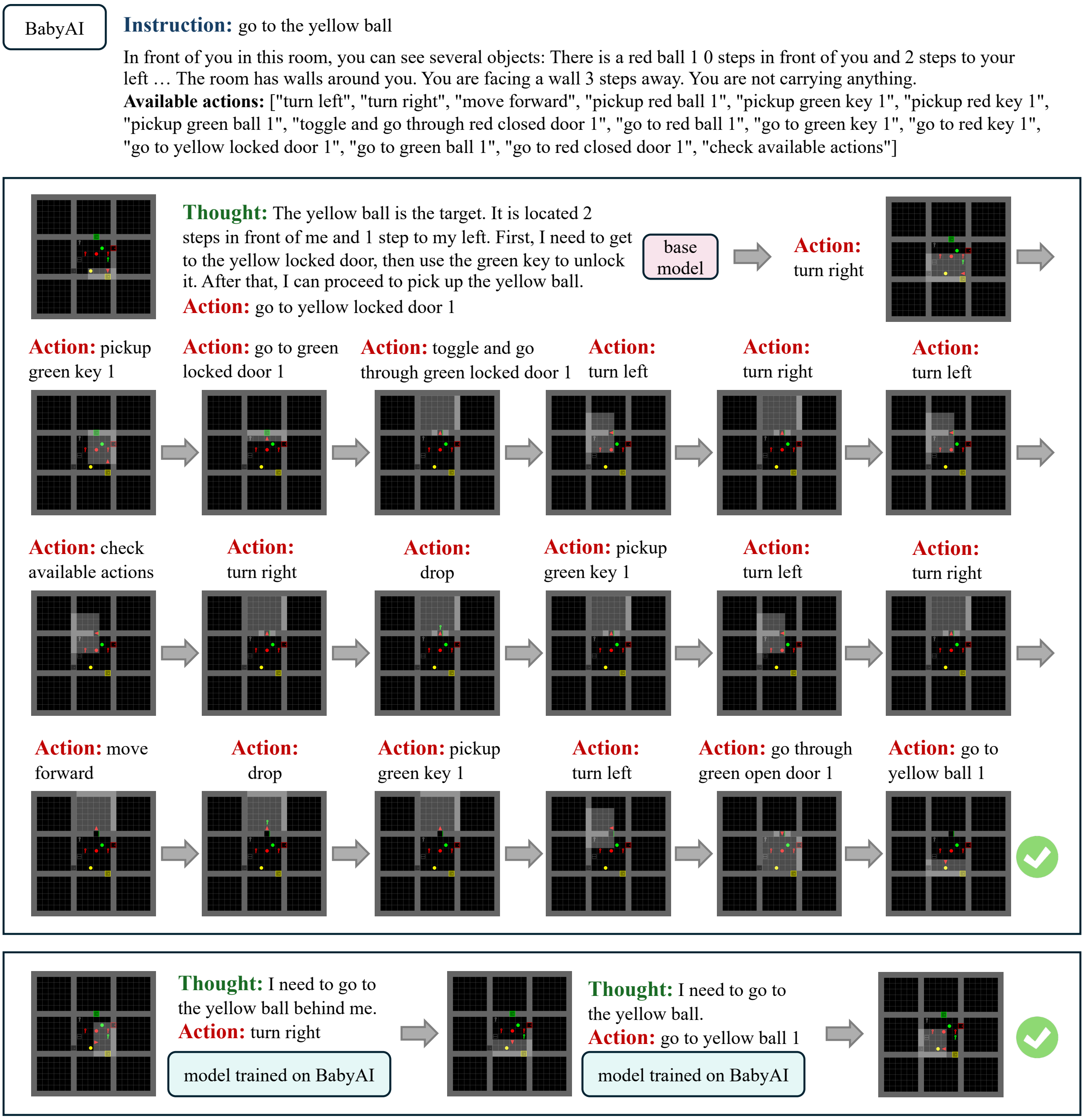

Figure 12 | A case study in the BabyAI environment. Compared to the base model, the trained model demonstrates significantly improved exploration efficiency.